Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
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Project Title:	Design and Validation of a Dynamic Exercise Countermeasure Device		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44195	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hoadley, David (Foster-Miller Inc.) Rajulu, Sudhakar (NSBRI) Yue, Guang (The Cleveland Clinic Foundation)		
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Exercise in space is a promising countermeasure to space flight-induced bone loss and muscle atrophy. The objective of this study was to develop a dynamic exercise countermeasure device that permits a jumping exercise in space. The device benefits multiple physiological systems - muscular, skeletal, neurovestibular and cardiovascular - without transmitting vibrations and impact forces to other parts of the spacecraft.
This three-year project was divided into three phases. In Year 1 we collaborated with Foster Miller Inc., a company that has expertise in the design of both lightweight structures and vibration isolation methodology, to construct a device that permits dynamic jumping exercise in microgravity and that is suitable for the International Space Station. In Year 2 we conducted ground-based studies in the exercise device to verify that muscle activation patterns are not compromised by the exercise device and that limb loading data, as measured by forces experienced under the feet during jumping, are in

Task Description:	the range that is believed to maintain the integrity of bone. Four different exercises were performed in the exercise device: 1) standard toe raises, 2) standard "squats", 3) "jumping rope" - in which consecutive mini-jumps were performed, and 4) maximum jumps - in which a series of maximum "vertical" jumps were performed. Test results are very promising in that we have been able to demonstrate that subjects exercising in the DECD experienced ground reaction forces ranging from 2-3 times bodyweight. These forces are greater than what is currently elicited by astronauts exercising in space and jumping exercises may, therefore, prove to be a more effective countermeasure to spaceflight-induced deconditioning. In Year 3 we confirmed the efficacy of the exercise device in true microgravity through experiments on the KC-135. Ground reaction forces were found to be at levels of high enough impact to provide an adequate daily loading stimulus to alleviate bone loss. With regards to muscle activity, results suggest that exercises in the DECD will stimulate higher muscular force requirements of certain muscle groups to better maintain overall muscle mass of the legs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefits of exercise the cardiovascular and musculoskeletal systems are widely accepted on Earth. Bone demineralization (bone mass loss) is a well-documented physiologic effect of long-duration spaceflight and as a consequence of disease and aging. Animal experiments on Earth have clearly indicated that: (1) certain bone strains and strain rates stimulate bone deposition, and (2) repetitive loading of the lower extremity can increase osteonal bone formation even as proximally as the vertebral column. Such studies have also indicated that a relatively small number of appropriate weight-loading cycles may be sufficient to stimulate bone deposition. By extending the philosophy of designing exercise modalities that elicit high impact forces, a dynamic exercise countermeasure device (DECD) that utilizes jumping as the mode of exercise will benefit a large population on Earth. This includes developing exercise protocols for patients with osteoporosis. Many such exercise plans are used; however, jumping may provide a more efficient and less time consuming way to address issues decreasing bone loss. An added benefit of the DECD is that it may aid patients with musculoskeletal disorders or traumatic injuries, which require rehabilitation therapy to strength and increase muscle mass to return to full function. This is due, in part to substantial activity found in lower extremity muscles during the resistive exercise such as squats and toe raises. In this manner, the DECD could act in the same manner as gym/therapy equipment.
Task Progress:	Fourteen subjects (7 males, 7 females) were tested in the KC135 aircraft in the DECD. Subject were passed physical examinations and attended physiological and altitude training necessary for all personnel flying on the KC135. All subjects performed each of the four different exercises during for a period of at least two parabolas. These exercised included: 1) standard toe raises, 2) standard "squats", 3) "jumping rope" - in which consecutive mini-jumps were performed, and 4) maximum jumps - in which a series of maximum "vertical" jumps were performed. Data were collected for each parabola and included head acceleration, ground reaction forces and electromyography of the gastrocnemius, tibialis anterior, vastus medialis and the biceps femoris of the left leg. Data were collected at intensity levels of 50 and 65% normal body weight. Maximal voluntary contraction (MVC) was recorded under non micro-gravity conditions. EMG data were recorded (sampling rate: 1000 Hz; low-pass filter: 6 Hz) from the vastus medialis, biceps femoris, tibialis anterior, and gastrocnemius of the left leg. In all circumstances, excluding the tibialis anterior while performing calf raises, the mean of each exercise repetition performed at 65% normal body weight tension produced greater %MVC compared with the 50% body weight condition. The overall increase in muscle activation was 5.6%, ranging from a 0.85% to a 12.6% increase. These results revealed that under micro-gravity conditions, higher magnitudes of muscular force have a direct relationship to the tension settings in the exercise device. Study has been completed and papers are currently being submitted to peer-reviewed journals. The PI is applying for further funding to bring the research on the DECD to the next phase on it way to being implemented in space. This will include a bed rest study to test the effectiveness of the device on alleviating bone mineral loss and muscle atrophy. Popular press coverage: 1. "Volunteers set to toss their cookies in the name of science", written by Shasta Clark. Crain's Cleveland Business, 25, 8, page 4, Feb 23rd-29th, 2004. 2. "Set to blast off", written by Sue Hoffman. The Solon Times, 26 (25), Pages 1 and 4, February 12th, 2004. 3. "Flying high for NASA", written by Sue Hoffman. Chagrin Valley Times/Solon Times Section B Cover story. March 11th, 2004.
Bibliography Type:	Description: (Last Updated: 07/13/2006)
Articles in Peer-reviewed Journals	Davis BL, Rajulu S, Tobin L, Tajaddini A. "Muscle activation levels during exercise in microgravity in a dynamic exercise countermeasure device." Journal of Electromyography and Kinesiology. Ed. Solomonow M. in press, 2004. , Jan-2004
Articles in Peer-reviewed Journals	D'Andrea SE, Perusek GP, Rajulu S, Perry J, Davis BL. "Jumping in simulated and true microgravity: response to maximal efforts with three landing types." Aviat Space Environ Med. 2005 May;76(5):441-7. PMID: 15892541 , May-2005
Presentation	Davis, B. L. "Design and development of an exercise countermeasure device for possible long-term space missions " Jan-2004
Presentation	Davis, B. L., D' Andrea, S. E., Eucker, E. E., Hoadley, D. "Design and development of an exercise countermeasure device for possible long-term space missions " Apr-2004
Presentation	Eucker, E., D' Andrea, S. E., Parker, K. R., and Davis, B. L. "Testing and Validation of a Dynamic Exercise Countermeasure Device " Oct-2004